

Assessing pre-training bias in Health data and estimating its impact on machine learning algorithms

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Abstract.

1. Introduction

Machine learning (ML) is increasingly used in various domains, including Health, which assists in diagnosis, prognosis, and treatment [Chen et al. 2021]. However, ML algorithms can potentially introduce biases, leading to unfair and potentially harmful outcomes [Obermeyer et al. 2019, Juhn et al. 2022].

This work aims to measure the risk of bias in Health datasets using pre-training bias metrics. It assesses their impact on the performance of supervised ML algorithms, correlating the bias metrics to the overall models' performance. We analyze four pre-training bias metrics [Hardt et al. 2021]: Class Imbalance (CI), Kullback-Leibler (KL) Divergence, Kolmogorov-Smirnov (KS), and Conditional Demographic Disparity in Labels (CDDL).

1.1. Class Imbalance (CI)

CI measures the representation of different classes within a dataset. It ranges from -1 to 1, with values near zero indicating a balanced representation. Positive values indicate overrepresentation of the advantaged group, while negative values indicate overrepresentation of the disadvantaged group.

1.2. Kullback-Leibler (KL) Divergence

KL Divergence measures the divergence between the label distributions of two facets. It ranges from 0 to $+\infty$, with values near zero indicating similar distributions and higher values indicating more significant divergence.

1.3. Kolmogorov-Smirnov (KS)

KS measures the maximum divergence between labels in the distribution for different facets. It ranges from 0 to 1, with values near zero indicating evenly distributed labels and values near one indicating unbalanced labels.

1.4. Conditional Demographic Disparity in Labels (CDDL)

CDDL measures the proportion of negative outcomes in a specific facet of a dataset. It ranges from -1 to 1, with positive values indicating demographic disparity favoring the advantaged group and negative values indicating disparity favoring the disadvantaged group.

2. Related Work

Many libraries and open-source tools have been developed to address machine learning fairness; for example, the AIF 360 library [Bellamy et al. 2018] provides different techniques to identify and mitigate bias. Studies have shown that word embedding algorithms may encode marginalized populations differently, perpetuating social bias. For example, [Zhang et al. 2020] demonstrated that such models associated African Americans and Blacks with prison and Whites and Caucasians with hospitals in a fill-in-the-blank style course of action. Similarly, Fairness Measures [Zehlike et al. 2017] provides different ways of classifying fairness, bias, and discrimination for any *csv* dataset. Additionally, FairLens [Synthesized.io 2023] is a Python library that automatically discovers and measures bias in data, providing a report on the data quality.

AI models trained on health data can be biased due to protected attributes like ethnicity and race. Various studies have explored different approaches for evaluating and mitigating bias in these models. [Júnior et al. 2022] analyzed the COMPAS dataset on criminal recidivism and found that the classifier was more prone to predict recidivism for Black people, favoring other demographics. In [Noseworthy et al. 2020], the author recommended that AI tools report the model’s performance for all the protected attributes. The study in [Park et al. 2021] found that bias mitigation algorithms outperformed simply removing the protected attribute that might cause bias. [Mandhala et al. 2022] used pre-training bias metrics to evaluate bias in models trained on three datasets, finding that mitigation strategies can improve model performance for the unprivileged classes.

While numerous works have focused on defining and mitigating bias, we have identified a gap in works that assess the direct impact of pre-training bias on the performance of machine learning algorithms, specifically in the Health domain. In this work, we aim to fill this gap by analyzing the correlation between pre-training bias metrics and the performance of supervised ML algorithms.

3. Methodology

Our methodology involves the following steps:

1. Data Collection: We collected four datasets from different sources
2. Data Preprocessing: We preprocessed the datasets to remove missing values and encode categorical variables
3. Increase/Decrease Bias: We artificially increased and decreased bias in the datasets, creating highly unbalanced and equally balanced versions
4. Pre-training Metrics: We calculated the CI, KL Divergence, KS, and CDDL metrics for each dataset
5. Train Models: We trained supervised ML algorithms on the datasets and evaluated their performance using accuracy, precision, recall, and F1-score
6. Correlation Analysis: We analyzed the correlation between the pre-training bias metrics and the performance of the ML algorithms

The datasets and their source and protected attributes are shown in Table 1.

Source	Dataset Name	Protected Attributes
[Newman et al. 1998]	HeartDataset	Sex
[Maslej et al. 2022]	IntersectionalBiasDataset	Sex, Race
[Teboul 2023]	DiabetesDataset	Sex, Age, Education, Income
[Tasci et al. 2022]	GliomaDataset	Gender, Race

Table 1. Datasets used in this work

4. Experiments and Results

5. Conclusion

6. References

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